

2023-Ph-H1-Q22

Section: Electricity

Topic: Internal Resistance

A cell has internal resistance. What happens to the terminal potential difference (t.p.d.) as current increases?

**Answer: C**

(It decreases)

**Guidance for Students:**

As current increases, more voltage is lost across internal resistance:

$$\text{t.p.d.} = E - Ir$$

So t.p.d. **decreases** with increasing current.

**Revision Tips:**

- $E$  = emf,  $r$  = internal resistance,  $I$  = current
- Terminal PD is what's **delivered to the circuit**
- Gradient of a  $V-I$  graph =  $-r$